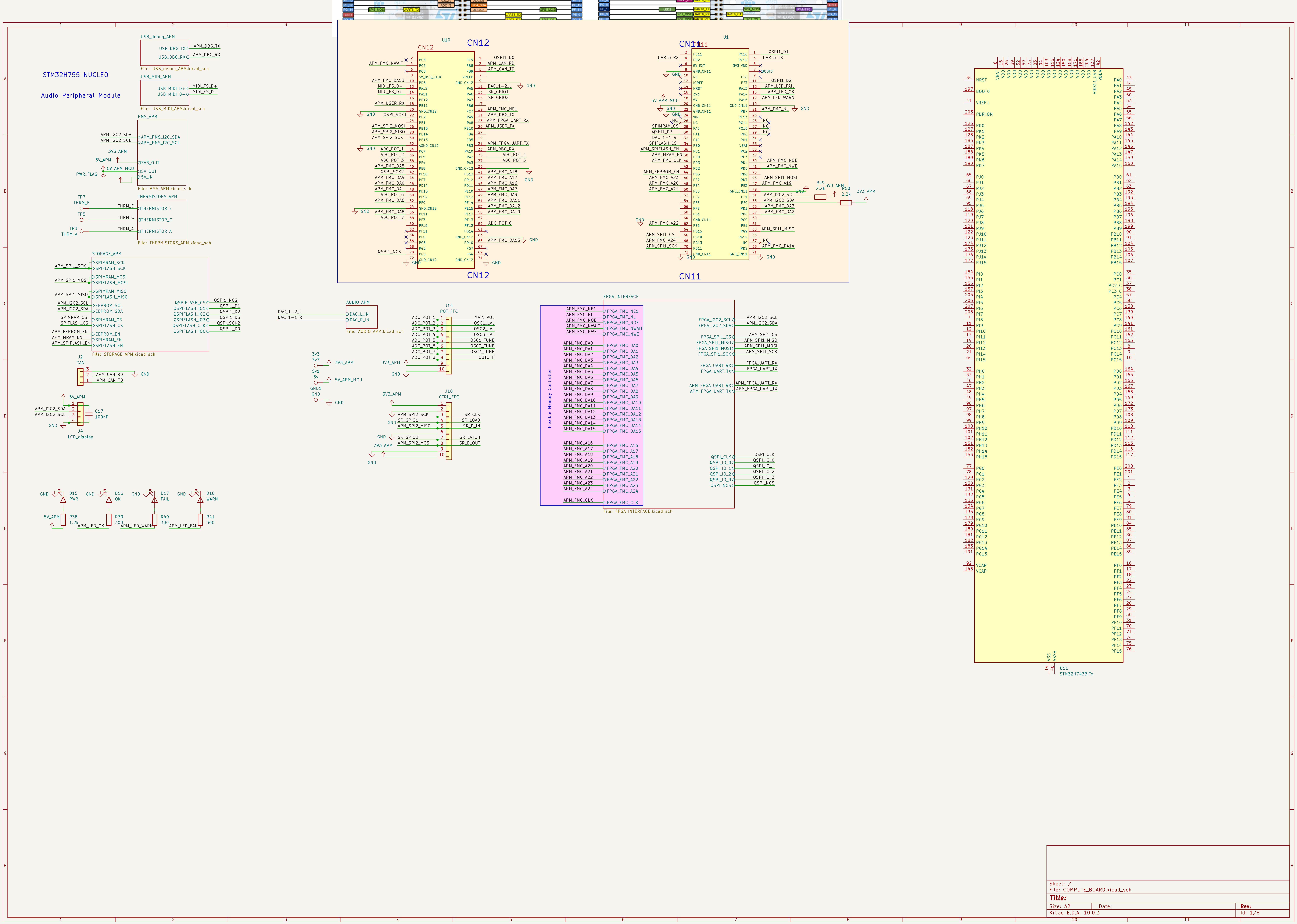
STM32H755 NUCLEO
Audio Peripheral Module
USB_debug_APM
File: USB_debug_APM.kicad_sch
USB_MIDI_APM
File: USB_MIDI_APM.kicad_sch
PMS_APM
File: PMS_APM.kicad_sch
THERMISTORS_APM
File: THERMISTORS_APM.kicad_sch
STORAGE_APM
File: STORAGE_APM.kicad_sch
AUDIO_APM
File: AUDIO_APM.kicad_sch
FPGA_INTERFACE
Flexible Memory Controller
File: FPGA_INTERFACE.kicad_sch
CN12
CN11
U10
U1
U11
STM32H743BITx
J2
CAN
J4
LCD_display
J14
POT_FFC
J18
CTRL_FFC
D15 PWR
D16 OK
D17 FAIL
D18 WARN
Sheet: /
File: COMPUTE_BOARD.kicad_sch
Title:
Size: A2
Date:
Rev:
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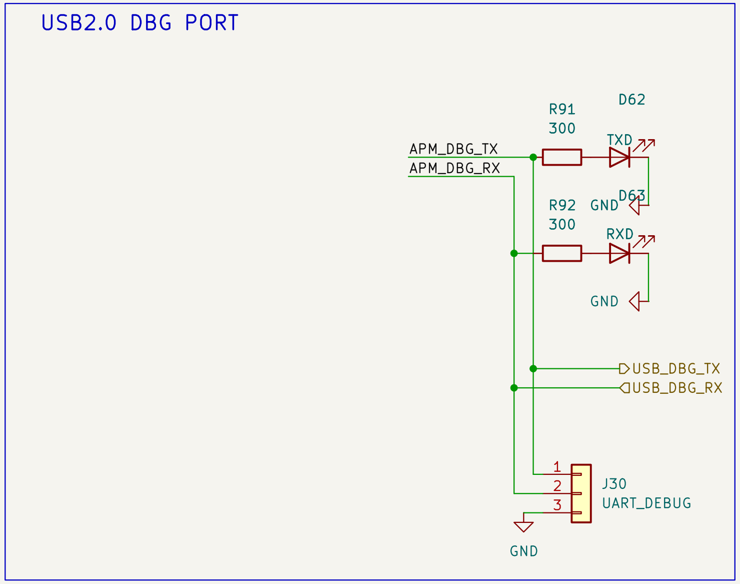
Id: 1/8

UART -> USB Debug Port

UART debug messages avaiable via a
USB port and a 3 pin jumper for
TX, RX, and GND UART connections



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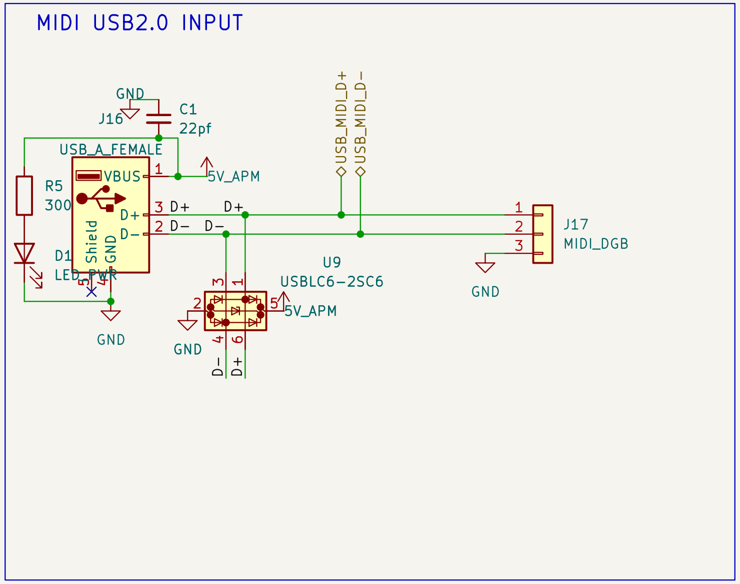
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 2/8



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File: USB_MIDI_APM.kicad_sch

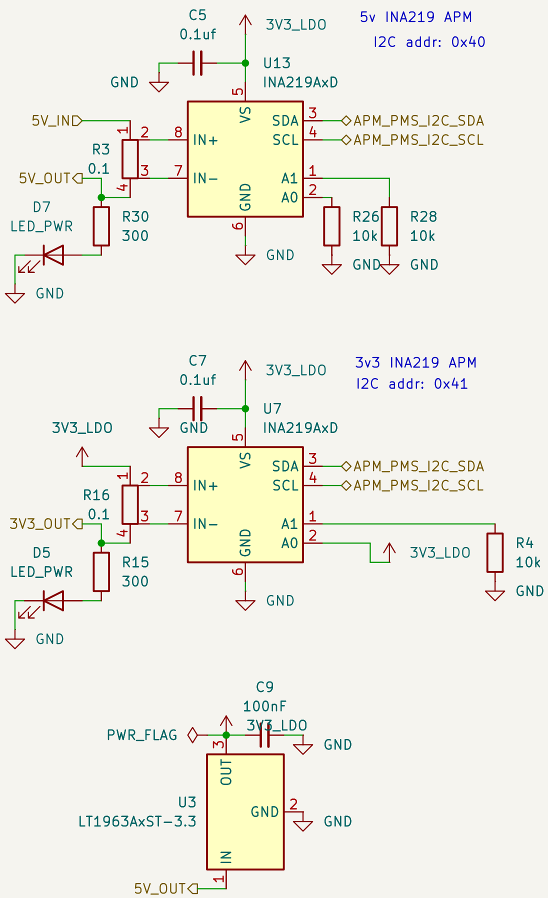
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 3/8

Power Monitor and Supply

PMS has an INA219 current sensor
for each board/node in the
PCB. Each takes in the 5V from it's
parent board and supplies 5V to said
board's circuit (its peripherals)
thus enabling monitoring



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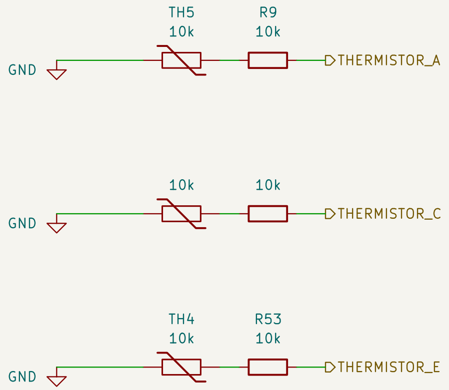
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 4/8



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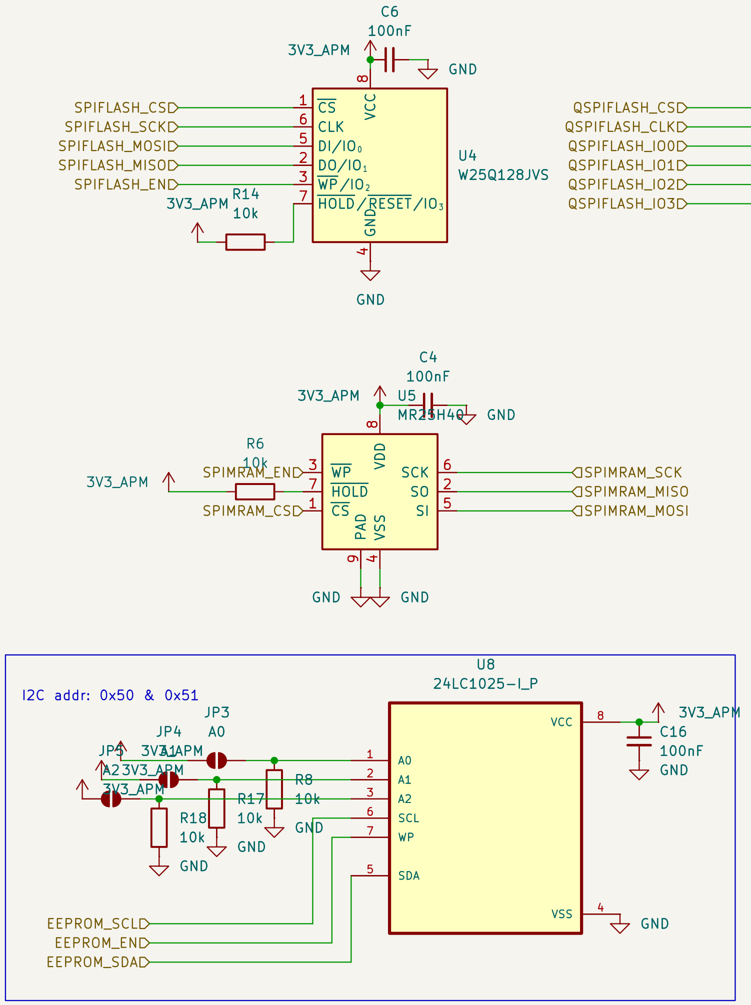
Title:

Size: A4 Date: **Rev:**
KiCad E.D.A. 10.0.3 Id: 5/8



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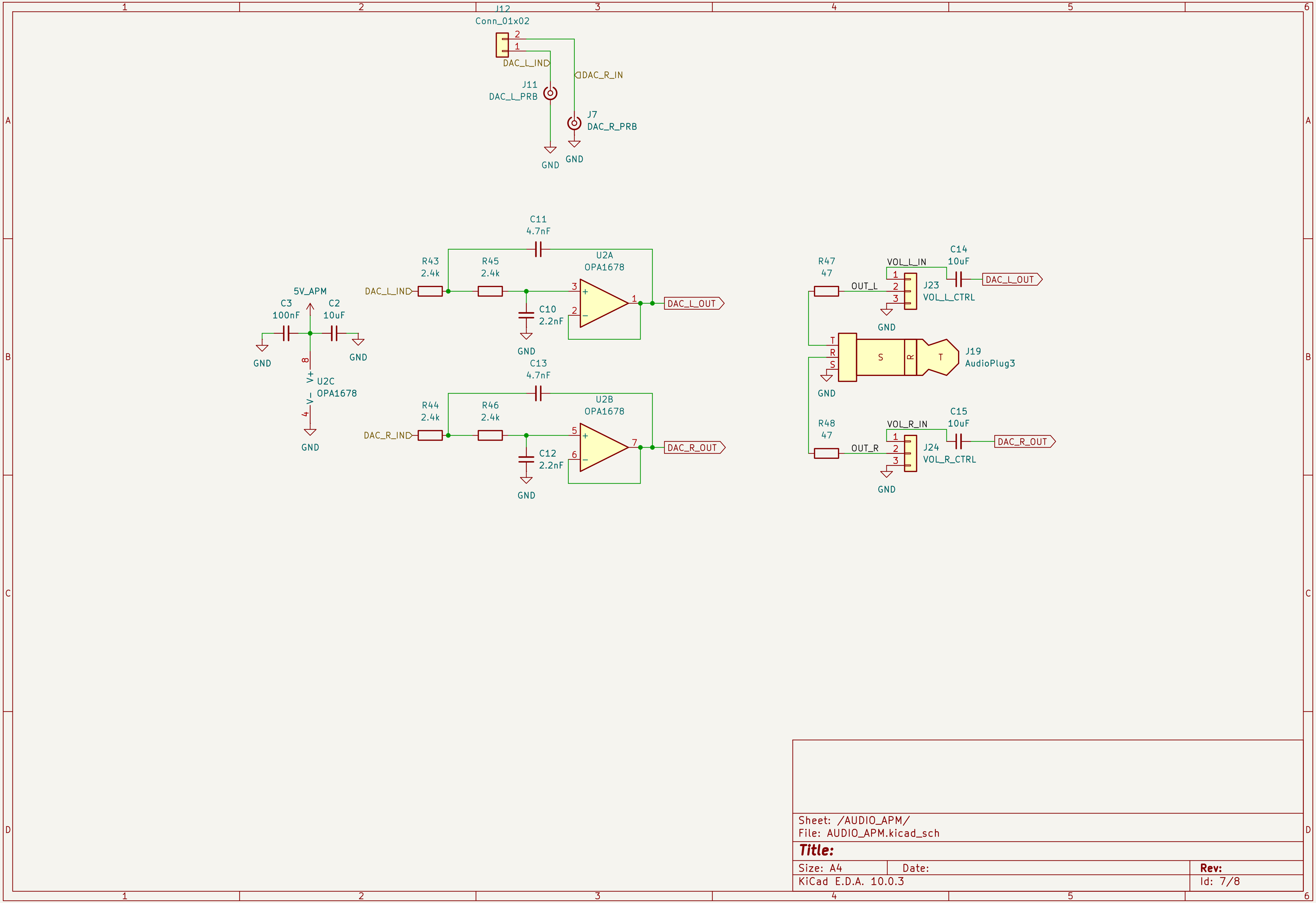
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 6/8

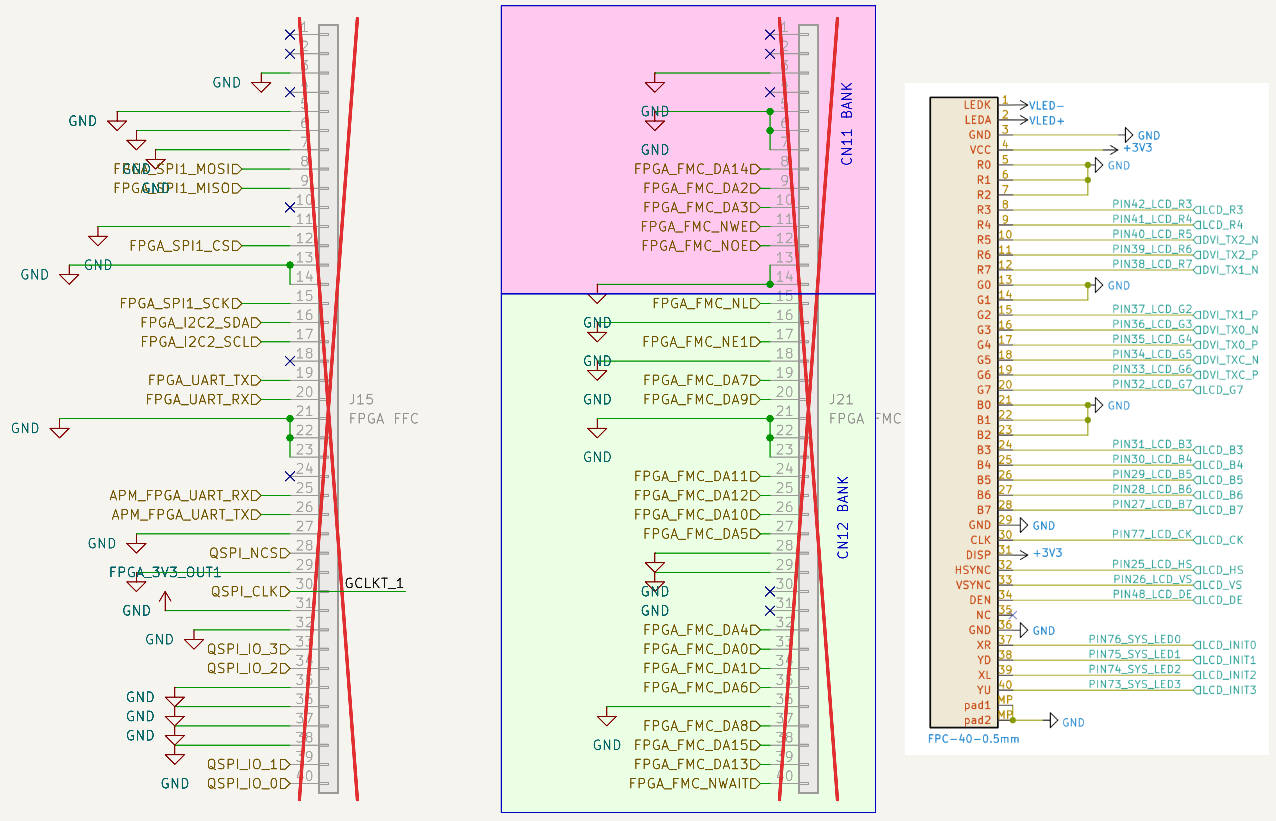

J12
Conn_01x02
DAC_L_IND
DAC_R_IN
J11
DAC_L_PRB
J7
DAC_R_PRB
GND
C11
4.7nF
R43
2.4k
R45
2.4k
U2A
OPA1678
DAC_L_IND
C10
2.2nF
DAC_L_OUT
5V_APM
C3
100nF
C2
10uF
U2C
OPA1678
C13
4.7nF
R44
2.4k
R46
2.4k
U2B
OPA1678
DAC_R_IND
C12
2.2nF
DAC_R_OUT
R47
47
OUT_L
VOL_L_IN
C14
10uF
J23
VOL_L_CTRL
J19
AudioPlug3
R48
47
OUT_R
VOL_R_IN
C15
10uF
J24
VOL_R_CTRL
Sheet: /AUDIO_APM/
File: AUDIO_APM.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 7/8

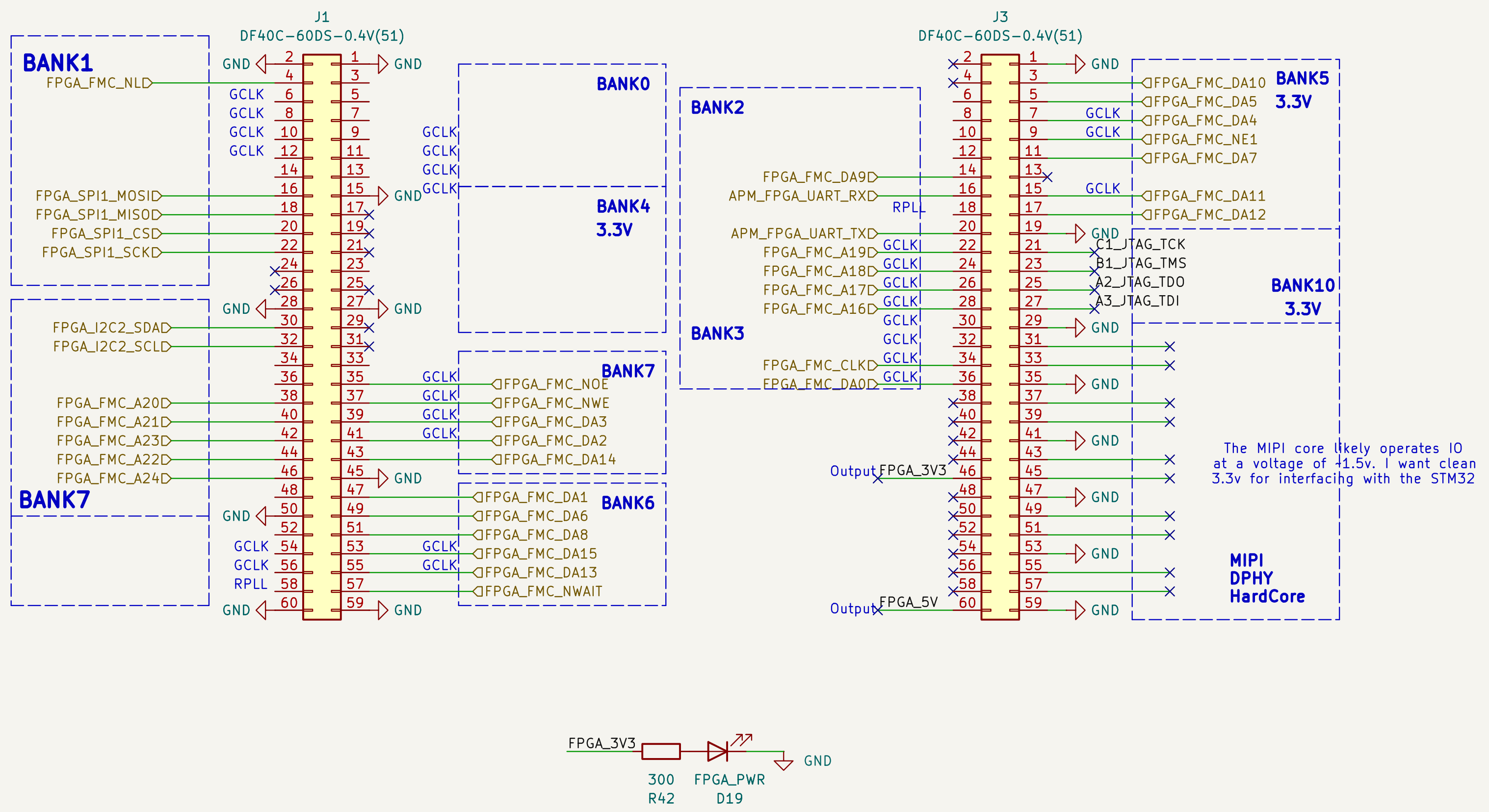
ACM FPGA Interface

Dual FPC STM32<->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with:
  - 16 bit muxed data
  - and async FMC

configuration resulting in a 64MB shared address space between the STM32 & FPGA





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File: FPGA_INTERFACE.kicad_sch

Title:

Size: A3 | Date: | Rev:

KiCad E.D.A. 10.0.3 | Id: 8/8